

Ram Milan Kumar Verma

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About Me

I am a Ph.D. research scholar in Aerospace Engineering at IIT Bombay with a strong academic foundation and industry experience in control systems. I was born in a small village near Bodhgaya, Gaya, Bihar. I am fortunate to have finished schooling from the esteemed Jawahar Navodaya Vidyalaya. During my undergraduate, I undertook multiple research projects in satellite attitude control, SLAM, and UAV systems. Passionate about solving real-world problems through robust mathematical modeling, simulation, and experimentation, I bring a multidisciplinary perspective backed by strong programming skills (Python, MATLAB, ROS, C++) and a deep interest in autonomous systems and intelligent control.

Education

Ph.D.	Indian Institute of Technology Bombay , <i>Aerospace Engineering</i>	Jul'23 – Present
	• Thesis topic: Guidance and Control of Autonomous Vehicles	
UG	Indian Institute of Technology Bombay	Jul'15 – Jun'20
	• Major: <i>Aerospace Engineering</i> Minor: <i>Electrical Engineering</i>	
	• Degree: Dual Degree (B.Tech + M.Tech.)	
	• Bachelor's Thesis: Decentralized Fusion algorithms for Multisensor Systems	
	• Master's Thesis: Distributed Filters with Directed Energy Weapon Systems	

Work Experience

Larsen & Toubro Precision Engineering and Systems <i>Flight Simulation and Control Engineer</i>	Powai, India
- Worked on developing the mathematical model and control system design of Manoeuvrable Expendable Aerial Target (MEAT) aircraft and its variants - Developed a mathematical model of tilt rotor aircraft similar to V-22 Osprey using MATLAB® - Responsible for making the models and control system deployable to the actual Flight Control Computer (FCC) and performing HILS testing and debugging - Simulated the swarming behaviour of aerial robots and implemented it on ground robots - Developed mathematical model for making 155mm artillery shells, a guided ammunition	Nov'22 – Jul'23
Sleepiz India <i>Digital Signal Processing Engineer</i>	Pune, India
Contactless medical grade sleep monitoring	Jul'20–Nov'22
- Collaboratively coded algorithms to estimate vital signs such as breathing rate and heart rate from raw data coming from Continuous Wave Doppler radar - Built learning-based models to assess the signal quality, classify sleep patterns, and measure parameters like Apnoe-Hypopnea-Index (AHI)	
Student Satellite Team, IIT Bombay <i>ADCS Subsystem Lead</i>	Powai, India
- Headed an interdisciplinary team of 10 members and helped devise the quality assurance practices - Recruited 6 students for the subsystem from 40 applicants evaluating their technical ability, approach, and teamwork - Contributed to Satellite 101 wiki, a compilation of comprehensive knowledge of satellite project	Feb'18 – Nov'18

Teaching and Mentorship Experience

Indian Institute of Technology Bombay <i>Teaching Assistant</i>	Powai, India
	Jul. 2023 to

- Help prepare lectures, assignments, projects and conduct tutorial, evaluating answersheets, projects and doubt clearing sessions for undergraduate and postgraduate students Present

• Courses:

- ★ **NPTEL Course:** Introduction to Autonomous Systems [*Ongoing*]
- ★ AE686: Guidance of Aerospace Vehicles [*Spring 2025*]
- ★ AE700: Guidance & Control of Unmanned Autonomous Vehicles [*Spring 2024*]
- ★ AE410: Navigation and Guidance [*Summer 2024*]
- ★ AE407: Modelling and Simulation [*Autumn 2023 and 2024*]
- ★ AE305/717: Flight Mechanics [*Spring 2020*]
- ★ AE695: State Space Methods for Flight Vehicles [*Autumn 2019*]

Indian Institute of Technology Bombay | *Teaching Assistant*

Navigation, Guidance, and Control of Unmanned Aerial Systems | Faculty Updation Program

Powai, India

Jul. 2023 to

Present

- Simulated path following algorithm for ground robots in MATLAB, Python and ROS-Gazebo followed by testing on actual robot (JetBots).
- Conducted hands on session for algorithm testing on ground robots, small drone (crazyFlie) and in ROS-Gazebo. The course was offered to faculties from various engineering institutes across India

Indian Institute of Space Science and Technology (IIIST) | *Teaching Assistant*

Workshop on Modeling and Simulation of UAS

Trivandrum,

India

Mar. 2025

- Prepared tutorials on mathematical modeling and simulation of quadcopter and fixed-wing aircraft using MATLAB®, SIMULINK, ROS-Gazebo and FlightGear
- Taught the participants the fundamental concepts of ROS and its use in their simulations

Publications

Journal Articles

- [J4] **Autonomous Docking of Unmanned Surface Vessel with Stationary and Moving Ships**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, [[Under preparation](#)]
- [J3] **True-Proportional-Navigation based Time Constrained Guidance of Unmanned Surface Vessel**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, [[Under preparation](#)]
- [J2] **Trajectory tracking control of surface vessels with constrained input magnitude and rate in the presence of disturbances**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, [[Under preparation](#)]
- [J1] **Barrier Lyapunov Function-based Controller for Surface Vessels with Motion Constraints**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, [[Under preparation](#)]

Conference Proceedings

- [C6] **True-Proportional-Navigation based Time Constrained Guidance of Unmanned Surface Vessel**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, *Proceedings of American Control Conference*, Denver, USA, 2025, [[Accepted for Publication](#)]
- [C5] **Approach Angle Constrained Guidance Strategy of Autonomous Surface Vessel**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, *Proceedings of American Control Conference*, Denver, USA, 2025, [[Accepted for Publication](#)]
- [C4] **Leader-Driven Formation Control for Unmanned Ground Vehicles: Simulation and Practical Implementation**
Ram Milan Kumar Verma, Swati Singh, Shashi Ranjan Kumar, Swati Singh, Hemendra Arya, *Tenth Indian Control Conference (ICC)*, Bhopal, India, 2024, pp. 160-165, [10.1109/ICC64753.2024.10883753](#) 
- [C3] **Trajectory Tracking Control of Surface Vessels With Constrained Input in the Presence of Disturbances**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, *Proceedings of AIAA Guidance, Navigation and Control Conference*, , Orlando, Florida, USA, 2025. [10.2514/6.2025-1904](#) 
- [C2] **Barrier Lyapunov Function-Based Controller for Surface Vessels With Motion Constraints**
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, *Proceedings of AIAA Guidance, Navigation and Control Conference*, , Orlando, Florida, USA, 2025. [10.2514/6.2025-1903](#) 

[C1] **Closed-Loop Simulation for Attitude Control of Nano-satellite**

Riya Singh, **Ram Milan Kumar Verma**, et al. , *In Advances in Small Satellite Technologies: Proceedings of 1st International Conference on Small Satellites (pp. 87-97)*. Springer Singapore. DOI: [10.1007/978-981-15-1724-2_9](https://doi.org/10.1007/978-981-15-1724-2_9)

Presentations

Posters

- [P2] **Iron Fish – Test bed for Unmanned Underwater Vehicles** Jul'24
Ram Milan Kumar Verma, Hemendra Arya, Shashi Ranjan Kumar, Presented at *Indo-German Workshop on Frontiers of Robot Learning (FRL 2024)*, IIT Bombay and *Cyber Physical Systems Symposium 2024*, IISc Bangalore
- [P1] **Barrier Lyapunov Function-based Controller for Surface Vessels with Motion Constraints** Jul'24
Ram Milan Kumar Verma, Shashi Ranjan Kumar, Hemendra Arya, Presented at *Indo-German Workshop on Frontiers of Robot Learning (FRL 2024)*, IIT Bombay

Symposiums

- [S1] **Trajectory tracking control of surface vessels with constrained input in the presence of disturbances**, **Ram Milan Kumar Verma**, Shashi Ranjan Kumar, Hemendra Arya, Presented at *Cyber Physical Systems Symposium 2024*, IISc Bangalore Jul'24

Industry Projects

- Vibration Tolerant Robust Controller Design for Gun Mounted on Naval Platform** [Ongoing]
Funding Agency: ARDE, Defence Research and Development Organisation (DRDO), India
- Wrote proposal and performed literature survey for gun barrel control mounted on a naval platform in the presence of disturbances
 - Developed a mathematical model, analyzed and designed various controllers such as PI, PID, LQG, LQR and Loop Shaping-based, to meet the performance specifications
- Engine Thrust Regulator for Ramjet Engine** [Aug'24]
Agency: DRDL, Defence Research and Development Organisation (DRDO), India
- Performed literature survey and wrote the proposal for thrust regulation of Ramjet engine

Proof-of-Concepts

- Control of Fully Actuated Quadrotor for Enhanced Precision Trajectory Tracking** [Ongoing]
Funding agency: Ministry of Electronics and Information Technology (MeitY), Government of India
- To develop a 6-DOF mathematical model of the fully actuated quadrotor and design controllers to perform specific tasks such as passing through narrow openings, crop monitoring, imaging, etc
- Designing of a UAV Capable to Maneuver in GPS-Denied Environment** [Ongoing]
Funding agency: Ministry of Electronics and Information Technology (MeitY), Government of India
- To design and implement a UAV capable of stabilizing itself after a collision in confined spaces
- Object Transportation Using Cooperative Unmanned Aerial Vehicles** ['24]
Funding agency: Ministry of Electronics and Information Technology (MeitY), Government of India
- Developed robust guidance and control schemes for a team of unmanned aerial vehicles (UAVs), enabling them to transport a load cooperatively from one location to another
- Obstacle Avoidance for an UAV Using Collision Cone Approach** ['24]
Funding agency: Ministry of Electronics and Information Technology (MeitY), Government of India
- Designed a controller for a quadrotor to ensure collision-free navigation in the presence of obstacles by utilising the collision cone approach

Additional Experience And Awards

- **ACC Student Travel Grant:** Awarded by American Automatic Control Council, USA [*'25*]
- **MHRD Postgraduate Fellowship:** for M.Tech. [*'20*] and Ph.D [*'23-ongoing*]
- Awarded Merit-cum-Means (**MCM**) Scholarship
- **Workshop:** Attended a program on “Automated Drones for Commercial Application” at SIT, Pune, India [*Dec'23*]
- **MRS 2023:** Participated in the Marine Robotics School 2023 conducted at CSIR-NIO, Goa, India [*Nov'23*]
- **Reviewer:** American Control Conference (ACC), IEEE World Forum on Internet of Things (IEEE WF-IoT)
- **Member:** Institute of Electrical and Electronics Engineers (IEEE), American Institute of Aeronautics and Astronautics (AIAA)
- Received a medal for being among the top runners in the General Championship Crossy [*Mar'24 and Apr'25*]
- Awarded *Bronze medal* in Post Graduate General Championship Crossy [*Jul'24*]
- Completed IIT Bombay Half Marathon [*Oct'24*]
- **FFE Award:** Recipient of Foundation For Excellence (FFE) scholarship award [*'18-'20*]

Skills

Programming: MATLAB®, SIMULINK, Git, Python, C++, $\text{\LaTeX}$, Java, R, SQL, HTML

Software: Visual Studio Code, ROS-Gazebo, Maple, Tableau

References

Kumar, Shashi Ranjan, Prof. Dr.

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